

Cosmic-ray electrons as ionizing agents of the intergalactic medium during the epoch of reionization

Draft manuscript

Abstract

Cosmic-ray (CR) electrons accelerated by the first stellar populations cool in an intergalactic medium (IGM) whose density, ionization state and radiation background evolve on comparable cosmological timescales. We follow electrons injected at $z_i = 10$ and 20 down to $z = 5.5$, integrating adiabatic, synchrotron, inverse-Compton (IC), Coulomb, excitation, ionization and bremsstrahlung losses simultaneously. The electron-impact cascade is closed with an energy-dependent binary-encounter-dipole secondary spectrum. We then integrate the differential Klein–Nishina spectrum of CMB photons emitted by every cascade electron. In the approved deposition estimate, each photon from 13.6 eV to 1 keV produces one H photoionization and a photoelectron of energy $E_\gamma - 13.6$ eV, which launches the same electron cascade. Direct collisions alone yield a minimum $W = K_{\text{ini}}/Y \simeq 35.4$ eV near 1 keV and saturate at $Y \simeq (0.8\text{--}1.2) \times 10^4$. The photon branch opens at a few MeV and creates a second efficient window near $K_{\text{ini}} \sim (5\text{--}9) \times 10^7$ eV, where $W \simeq 39$ eV. It raises the high-energy plateau to 2.9×10^6 ($z_i = 20$) and 4.0×10^6 ($z_i = 10$). Replacing 1 keV by the redshift-dependent interaction-horizon ceiling raises these values to 3.9×10^6 and 4.5×10^6 . Thus leptonic ionization has two productive energy windows, while its global importance remains controlled by the injection spectrum, electron escape and photon radiative transfer.

1 Introduction

1.1 Reionization and cosmic dawn

Hydrogen reionization is the last global phase transition of the baryonic Universe. The Ly α forest indicates that reionization-related fluctuations persist to at least $z \simeq 5.3$, whereas the CMB optical depth $\tau = 0.054 \pm 0.007$ places the midpoint of a rapid transition near $z \simeq 7.7$ and limits a substantial ionized fraction at $z \gtrsim 15$ [Bosman et al., 2022, Planck Collaboration et al., 2020]. The range $5.5 \leq z \leq 20$ therefore brackets the transition and the preceding cosmic-dawn interval in which the first sources began to alter the IGM [Robertson, 2022, Klessen and Glover, 2023].

The ionizing-photon budget is still uncertain. Early galaxies are the standard source population, but their inferred abundance, ionizing efficiency and escape fraction can overproduce the ionization history allowed by the CMB and Ly α forest, a tension often called the post-JWST photon-budget crisis [Robertson, 2022, Muñoz et al.,

2024]. Independent 21-cm limits also disfavor models in which the neutral IGM remains at the adiabatic cooling floor until late times, implying energy deposition before ionized regions overlap [HERA Collaboration et al., 2023]. Heating and ionization by fast particles are two outcomes of the same microscopic energy-loss problem [Furlanetto and Stoever, 2010].

1.2 Why cosmic-ray electrons?

The first stellar populations provide shocks and compact remnants capable of particle acceleration. Metal-free stars form in early minihaloes, and their winds, supernovae and remnants inject mechanical power on timescales short compared with the duration of reionization [Klessen and Glover, 2023, Owen et al., 2017]. Collisionless quasi-parallel shocks accelerate ions and electrons into non-thermal power laws, while low metallicity enhances the output of high-mass X-ray binaries that may also drive non-thermal outflows [Park et al., 2015, Kaur et al., 2022]. Whether these electrons reach the diffuse IGM is a separate escape problem; throughout this paper “injected electron” means one that has already escaped its host.

CR heating of the early IGM has been studied primarily for protons. Primordial supernovae can supply sub-relativistic CRs that heat neutral gas, and cosmological transport calculations find temperature increments of order 10–200 K by $z \sim 10$ for plausible hadronic spectra, while their direct ionization contribution remains small [Sazonov and Sunyaev, 2015, Leite et al., 2017]. The spatial structure of this heating depends on diffusion and magnetic self-confinement [Jana and Nath, 2018, Gessey-Jones et al., 2023]. Electrons require a separate calculation because IC losses on the CMB are efficient and create secondary photons that have no hadronic analogue at fixed particle energy [Blumenthal and Gould, 1970, Slatyer et al., 2009].

We calculate the ionization yield per escaped electron before convolving it with any source spectrum. This ordering separates atomic and cosmological physics from uncertain acceleration efficiencies. It also corrects a limitation of the earlier draft: photon-mediated ionizations are now computed from the differential IC spectrum rather than inferred from an energy-loss curve.

2 Physical model

2.1 Fiducial IGM

We adopt the approved homogeneous, pure-hydrogen baseline

$$n_{\text{HI}}(z) = 1.8 \times 10^3 \left(\frac{1+z}{21} \right)^3 \text{ m}^{-3}, \quad x_e \equiv \frac{n_e}{n_{\text{HI}}} = 10^{-4}, \quad (1)$$

and a frozen-in magnetic field $B(z) = B_0(1+z)^2$ with $B_0 = 1 \text{ nG}$. Cosmological ages and $H(z)$ use the Planck-2018 cosmology as implemented in `astropy` [Planck Collaboration et al., 2020, Astropy Collaboration et al., 2022]. The ages at $z = 20, 10$ and 5.5 are 178.1, 471.4 and 1038.5 Myr, respectively. Helium and recombinations are omitted, and the residual ionized fraction is held fixed.

The CMB and magnetic energy densities both scale as $(1+z)^4$ under these assumptions. In the Thomson limit their ratio is redshift-independent,

$$\frac{U_B}{U_{\text{CMB}}} = \frac{B_0^2}{2\mu_0 U_{\text{CMB},0}} = 9.5 \times 10^{-8}, \quad (2)$$

and equality would require $B_0 \simeq 3.2 \mu\text{G}$. Klein–Nishina suppression raises the synchrotron-to-IC ratio at the highest energies, but the numerical loss model shows that synchrotron cooling remains dynamically negligible over the range studied here [Blumenthal and Gould, 1970].

2.2 Seven simultaneous electron losses

For kinetic energy K_e , total energy $E = K_e + m_e c^2$ and $p^2 c^2 = K_e(K_e + 2m_e c^2)$, the trajectory is

$$\frac{dK_e}{dt} = - \sum_{j=1}^7 L_j(z, K_e). \quad (3)$$

Adiabatic redshifting gives $L_{\text{ad}} = H(z)p^2 c^2/E$. The isotropic-pitch-angle synchrotron and IC rates are

$$L_{\text{rad}} = \frac{4}{3} \sigma_T c U \frac{p^2 c^2}{(m_e c^2)^2}, \quad U = U_B \text{ or } U_{\text{CMB}} F_{\text{KN}}(b), \quad (4)$$

where $b = 4\gamma k T_{\text{CMB}}/(m_e c^2)$ and F_{KN} is obtained by integrating the Klein–Nishina kernel over the Planck spectrum [Blumenthal and Gould, 1970]. Coulomb losses on the residual plasma follow Gould [1972]. Excitation is summed over $\text{H}(1s \rightarrow np)$ for $n = 2\text{--}10$ using scaled plane-wave-Born cross-sections [Stone et al., 2002]. Collisional ionization uses the relativistic binary-encounter-Bethe (RBEB) cross-section, with the binding energy plus the mean secondary-electron energy removed per event [Kim et al., 2000, Kim and Rudd, 1994]. Bremsstrahlung uses the screened and unscreened limits of Blumenthal and Gould [1970].

The seven rates are integrated as one stiff ordinary differential equation with an implicit Radau method. A terminal event stops a trajectory at $K_e = 3kT_{\text{CMB}}(z)/2$.

Table 1: Computed regime boundaries. $E_{\gamma, \text{hor}}$ is the photon energy at which the adopted interaction mean free path equals $c/H(z)$; all other columns are electron kinetic energies in eV.

z	K_2	K_3	$E_{\gamma, \text{hor}}$	K_4
20.0	7.60×10^4	4.20×10^6	1820	3.25×10^8
15.0	9.32×10^4	4.88×10^6	1550	3.43×10^8
10.0	1.23×10^5	5.99×10^6	1272	3.75×10^8
8.0	1.43×10^5	6.67×10^6	1151	3.94×10^8
5.5	1.82×10^5	7.93×10^6	983	4.29×10^8

Seven augmented states simultaneously accumulate $\int L_j dt$, ensuring that the channel fractions are evaluated on the same trajectory and sum to the kinetic energy lost. Electrons are injected at $z_i = 10$ or 20 and followed to $z = 5.5$ if they have not thermalized earlier.

3 Structure of the cooling problem

3.1 Instantaneous dominance and regime boundaries

Figure 1 shows three broad electron-loss domains. Coulomb and atomic losses control sub-relativistic particles, IC losses rise rapidly above the collisional domain, and adiabatic, synchrotron and bremsstrahlung terms do not dominate within the plotted parameter range. Instantaneous dominance does not itself determine deposition: a cooling electron traverses several domains before thermalization.

Four useful boundaries organize the (z, K_e) plane. $K_1 = 13.6057 \text{ eV}$ is the H ionization threshold. K_2 solves $L_{\text{ion}}(z, K_2) = L_{\text{IC}}(z, K_2)$ and decreases with redshift because atomic losses scale approximately as $(1+z)^3$ whereas the CMB energy density scales as $(1+z)^4$ [Kim et al., 2000, Blumenthal and Gould, 1970]. K_3 is the electron energy for which an up-scattered photon from the high-energy CMB percentile first reaches 13.6 eV . K_4 maps the photon interaction-horizon energy, defined by $\lambda_\gamma = c/H$, back to a parent-electron energy using the IC kinematics [Blumenthal and Gould, 1970, Karzas and Latter, 1961].

The photon boundary is intrinsically broad because the CMB and IC output are spectra, not monoenergetic beams. We use K_3 only as a diagnostic; the photon yield itself is always integrated over the full differential spectrum. The fixed 1-keV absorbed band is the headline assumption approved for this work, while $E_{\gamma, \text{hor}}(z)$ is retained only as an opacity sensitivity test.

3.2 Integrated energy budgets

The accumulated channel fractions are more informative than an instantaneous map. At 100 eV , collisional ionization, excitation and Coulomb heating receive approximately 0.48, 0.40 and 0.12 of the initial energy. Near 1 keV the ionization fraction rises to about 0.68 and exci-

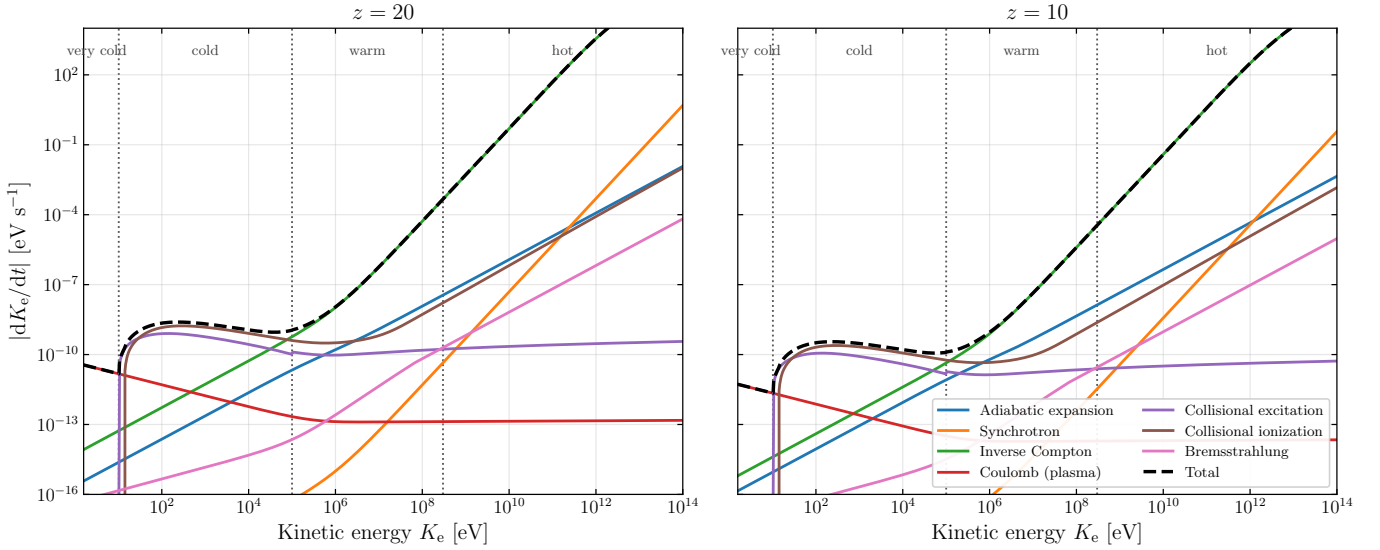


Figure 1: Instantaneous loss rates at $z = 20$ and 10 for the approved IGM. The three vertical dotted lines are the nominal conceptual divisions at 10.2 eV, 100 keV and 300 MeV; the physically computed boundaries are listed in Table 1. Every curve is evaluated from the common loss module using the cross-sections cited in Sect. 2.

tation receives 0.31. At 1 MeV IC already receives 0.80 ($z_i = 20$) or 0.66 ($z_i = 10$), and above 100 MeV it receives at least 0.99. These fractions are direct outputs of the simultaneous integration based on Blumenthal and Gould [1970], Gould [1972], Stone et al. [2002], Kim et al. [2000].

The distinction in the caption is essential. Counting the entire IC loss as locally deposited would overestimate ionization, whereas discarding it would miss the productive 13.6 eV–1 keV photons. The differential calculation below selects the relevant part of the radiated spectrum.

4 Ionization cascade

4.1 Electron-impact branch

An ionizing electron collision creates one ion and a secondary electron. The RBEB singly differential cross-section supplies a normalized, energy-dependent distribution $P_{\text{sec}}(\epsilon|K_e)$ [Kim and Rudd, 1994]. Along a parent trajectory the direct yield satisfies

$$\dot{Y}_e = n_{\text{HI}} v \sigma_{\text{ion}}(K_e) \left[1 + \int d\epsilon P_{\text{sec}}(\epsilon|K_e) Y_e(\epsilon, z) \right]. \quad (5)$$

Because $\epsilon \leq (K_e - B_{\text{H}})/2 < K_e$, the recursion is causal and can be marched upward on an energy grid. Each generation follows Eq. (3) on its own cosmological trajectory. At low energy the solution agrees with independent neutral-gas deposition calculations [Shull and van Steenberg, 1985, Furlanetto and Stoever, 2010].

4.2 Productive inverse-Compton photons

For an isotropic blackbody field, the differential IC rate is

$$\begin{aligned} \frac{d\dot{N}_\gamma}{dE_\gamma} &= \frac{3\sigma_T c}{4\gamma^2} \int d\epsilon \frac{n(\epsilon)}{\epsilon} G(q, \Gamma), \\ q &= \frac{E_\gamma}{\Gamma(\gamma m_e c^2 - E_\gamma)}, \quad \Gamma = \frac{4\gamma\epsilon}{m_e c^2}, \\ G &= 2q \ln q + (1 + 2q)(1 - q) + \frac{(\Gamma q)^2(1 - q)}{2(1 + \Gamma q)}, \end{aligned} \quad (6)$$

with $1/(4\gamma^2) \leq q \leq 1$ [Blumenthal and Gould, 1970]. Integrating Eq. (6) over all output energies reproduces the IC power used in Eq. (4) to 0.4 per cent at 10 MeV and to 0.2 per cent above 100 MeV. This independent energy check ensures consistency between the trajectory and photon-counting calculations.

Following the approved hypothesis, every photon with $B_{\text{H}} = 13.6057 \text{ eV} \leq E_\gamma \leq E_{\text{max}} = 1 \text{ keV}$ produces one H photoionization. Its photoelectron begins with $K_e = E_\gamma - B_{\text{H}}$ and launches the same electron cascade, so

$$\dot{Y}_\gamma = \int_{B_{\text{H}}}^{E_{\text{max}}} dE_\gamma \frac{d\dot{N}_\gamma}{dE_\gamma} [1 + Y_e(E_\gamma - B_{\text{H}}, z)]. \quad (7)$$

Equation (7) is evaluated for the primary and every collisional secondary. The total is $Y = Y_e + Y_\gamma$. Photoelectrons in the adopted band have $K_e < 1 \text{ keV}$, so they do not themselves generate ionizing IC photons; their descendants remain within the electron-impact cascade [Furlanetto and Stoever, 2010, Slatyer et al., 2009].

For sensitivity only, we replace 1 keV by the redshift-dependent energy where

$$\lambda_\gamma^{-1} = n_{\text{HI}} \sigma_{\text{pi}}(E_\gamma) + (n_{\text{HI}} + n_e) \sigma_{\text{KN}}(E_\gamma) = \frac{H(z)}{c}. \quad (8)$$

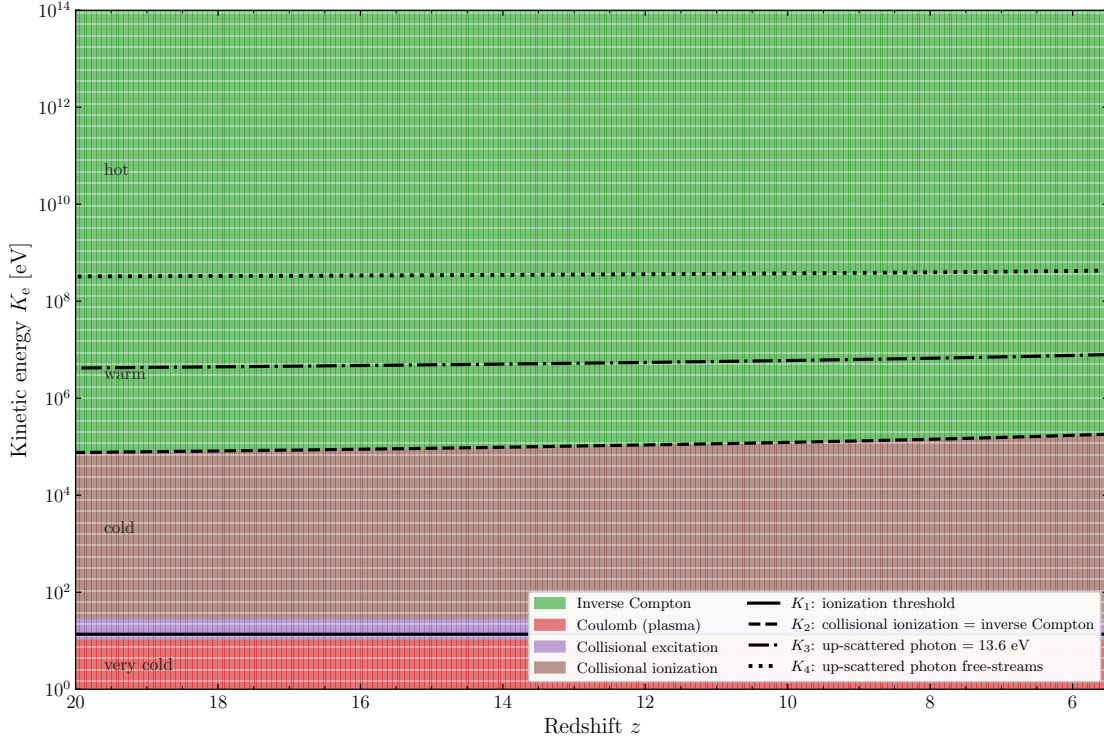


Figure 2: Dominant instantaneous loss channel over $5.5 \leq z \leq 20$, with the four diagnostic boundaries. The K_3 and K_4 curves describe the fate of IC photons rather than a change in the electron loss mechanism.

The H photoionization and Klein–Nishina cross-sections follow [Karzas and Latter \[1961\]](#), [Blumenthal and Gould \[1970\]](#). Equation (8) defines an interaction horizon, not a complete absorption probability, because scattering and cosmological redshifting require radiative transfer.

5 Ionization results

5.1 The direct collisional window

Below $\sim 10^5$ eV, atomic collisions control the yield. The effective energy per ionization has a broad minimum $W \simeq 35.4$ eV near 1 keV, reproducing the neutral-gas limit of [Shull and van Steenberg \[1985\]](#), [Furlanetto and Stoever \[2010\]](#). At 100 eV the cascade produces 2.65 ionizations per electron; at 1 keV it produces 28.3. The yield is nearly redshift-independent here because the cooling interval is much shorter than the cosmological evolution time.

Once IC cooling overtakes collisional ionization, less primary energy reaches the direct cascade. The electron-only result therefore flattens at 7.7×10^3 ionizations for $z_i = 20$ and 1.24×10^4 for $z_i = 10$. This plateau is the correct lower bound when emitted photons are not followed.

5.2 The IC-photon window

The IC spectrum develops an ionizing tail at a few MeV and dominates the ionization count by $K_{\text{ini}} \sim 10^7$ eV. At

$K_{\text{ini}} \simeq (5-9) \times 10^7$ eV the fixed-band calculation reaches a second minimum $W = 38.7-39.0$ eV. Thus the photon branch can be almost as efficient as the keV collisional cascade under the one-absorption prescription.

At $K_{\text{ini}} = 2.6 \times 10^8$ eV the fixed photon branch produces 2.8×10^6 ionizations for $z_i = 20$ and 3.8×10^6 for $z_i = 10$, whereas direct impacts contribute only 6.9×10^3 and 1.1×10^4 . At higher K_{ini} , added energy emerges mainly above the absorbed band and the yield saturates at 2.9×10^6 and 4.0×10^6 . A productive photon then gives 3.6 ionizations on average after its photoelectron cascade.

The interaction-horizon ceiling increases the plateaux to 3.9×10^6 ($z_i = 20$) and 4.5×10^6 ($z_i = 10$), a 37 and 13 per cent change. This sensitivity is modest compared with the two-to-three-order-of-magnitude difference between the photon and direct branches, but it is large enough that future work should replace both ceilings with radiative transfer [[Slatyer et al., 2009](#), [Slatyer, 2016](#)].

6 Cooling histories and spatial interpretation

Cooling accelerates strongly with redshift because both gas density and the CMB energy density increase. A 100 eV electron injected at $z_i = 10$ thermalizes after $\Delta z \simeq 0.002$, whereas 1 MeV and 1 TeV electrons reach the floor after $\Delta z \simeq 1.82$ and 2.85. At $z_i = 20$ the corresponding intervals are approximately 0.001, 0.89 and 1.23. The convergence of

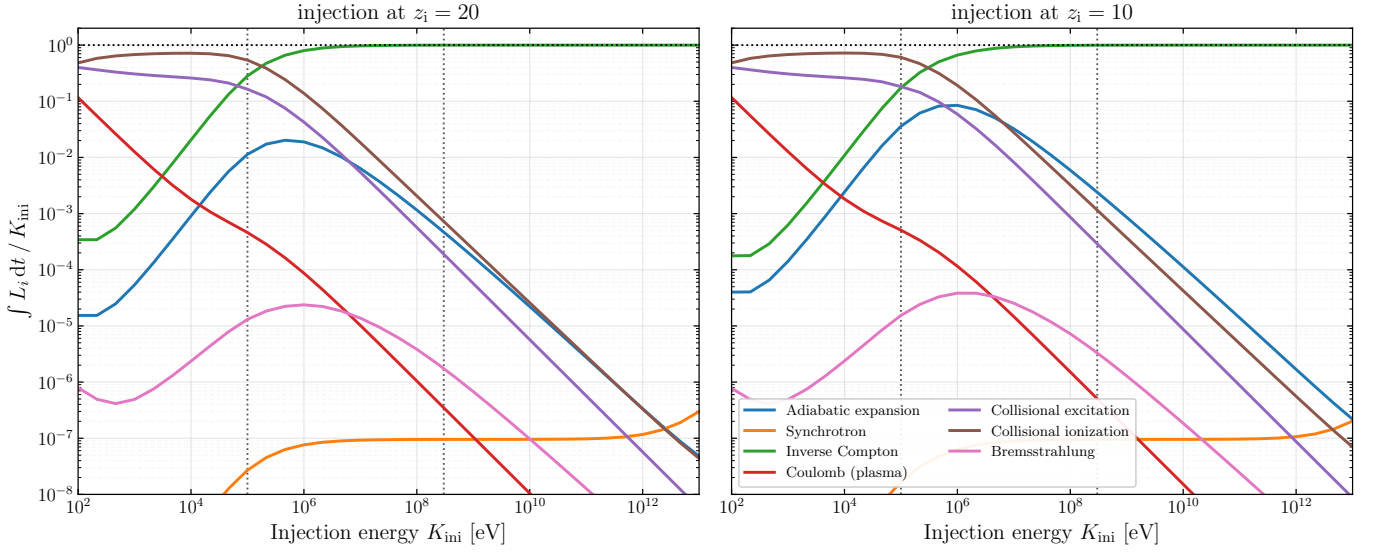


Figure 3: Fraction of the injection energy carried by each electron-loss channel before thermalization, for $z_i = 20$ and 10. IC energy in this figure is radiated energy; only the spectral fraction inside the adopted photon band is converted into ionizations in Sect. 4.

the high-energy histories reflects the IC-dominated descent into the same lower-energy cascade [Blumenthal and Gould, 1970].

The accumulated path length is not a shell radius. For example, an electron injected at $z_i = 20$ with 10^5 eV accumulates about 8 comoving Mpc before thermalization, a 1 MeV electron accumulates about 50 cMpc, and injection above 10^8 eV approaches 78 cMpc. These are upper limits on displacement. In the approved $B_0 = 1$ nG field the gyroradius is many orders of magnitude smaller, so displacement depends on field geometry, pitch-angle scattering and the diffusion coefficient [Jana and Nath, 2018, Ruszkowski and Pfrommer, 2023].

Consequently, the yield calculation predicts how much ionization is produced per electron but not where it is deposited. CR heating can be patchy for small diffusion coefficients and more uniform for faster diffusion, a distinction that affects 21-cm observables [Jana and Nath, 2018, Gessey-Jones et al., 2023]. A spatial source model must convolve the present yield kernel with a transport Green’s function rather than identify Fig. 5 with a bubble size.

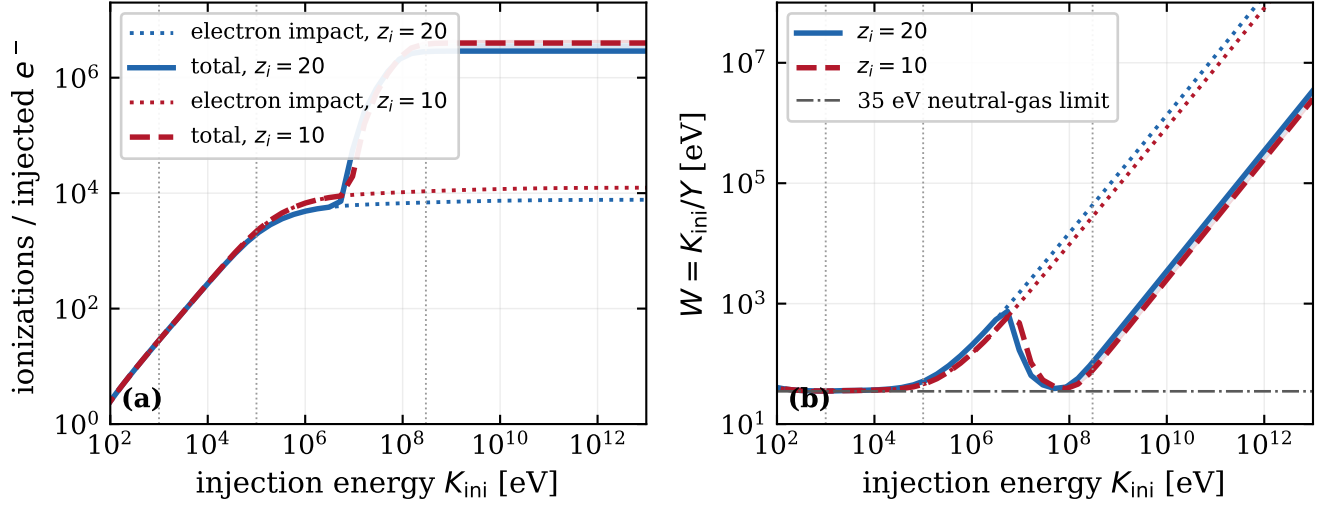


Figure 4: (a) Ionizations per injected electron. Dotted curves show the electron-impact cascade; thick curves add IC photons in the 13.6 eV–1 keV band and their photoelectron cascades. Shading extends the total to the redshift-dependent interaction-horizon ceiling. (b) Effective energy per ionization. The first minimum is the low-energy collisional cascade and the second is the IC-photon window.

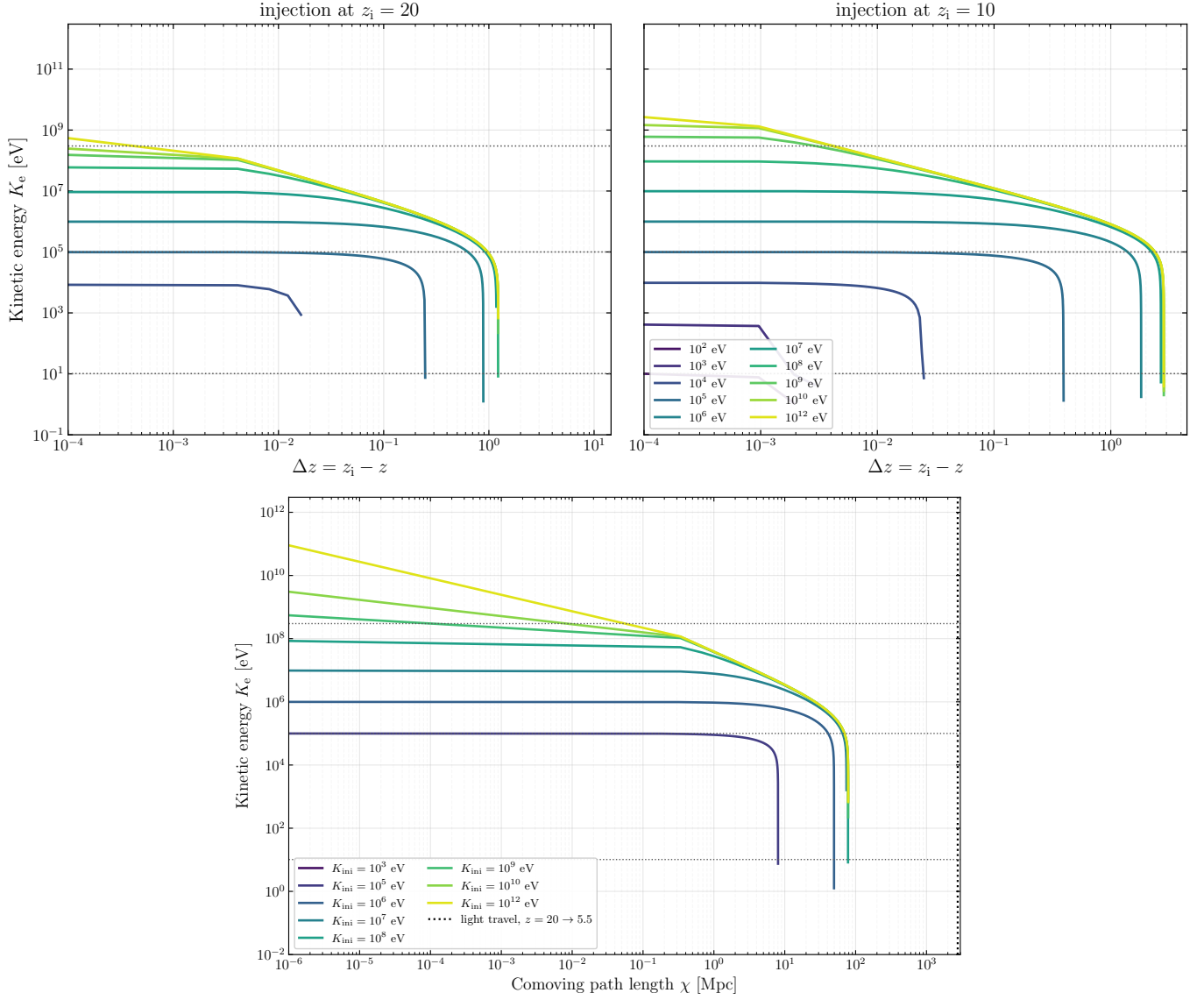


Figure 5: Top: electron kinetic energy against elapsed redshift interval for injection at $z_i = 20$ and 10; curves terminate at the thermal floor and high-energy histories converge after IC cooling. Bottom: energy against accumulated comoving path length for injection at $z_i = 20$. The travel length is not a radial displacement in a magnetized medium.

Table 2: Spectrum-averaged energy per ionization $\overline{W}$ [eV] for $100 \text{ eV} \leq K_{\text{ini}} \leq 1 \text{ TeV}$. “Direct” omits IC-photon ionizations, “1 keV” uses the approved fixed band, and “horizon” uses the redshift-dependent ceiling of Eq. (8).

p	$z_i = 20$			$z_i = 10$		
	Direct	1 keV	Horizon	Direct	1 keV	Horizon
1.8	888	290	263	804	271	260
2.0	107	79.8	77.4	102	77.1	76.2
2.2	46.3	44.2	44.0	45.3	43.5	43.4
2.5	37.8	37.7	37.7	37.6	37.5	37.5
3.0	37.4	37.4	37.4	37.4	37.4	37.4

7 Injection spectra and global scaling

For an illustrative escaped spectrum $dn/dK_{\text{ini}} = AK_{\text{ini}}^{-p}$ between $K_{\text{min}} = 100 \text{ eV}$ and $K_{\text{max}} = 1 \text{ TeV}$, the spectrum-averaged energy per ionization is

$$\overline{W}(p) = \frac{\int_{K_{\text{min}}}^{K_{\text{max}}} K_{\text{ini}}^{1-p} dK_{\text{ini}}}{\int_{K_{\text{min}}}^{K_{\text{max}}} K_{\text{ini}}^{-p} Y(K_{\text{ini}}) dK_{\text{ini}}}. \quad (9)$$

The normalization A cancels exactly. Power-law injection is motivated by diffusive shock acceleration, but the extrapolation to 100 eV is illustrative and should not be mistaken for a universal source spectrum [Park et al., 2015].

Table 2 shows that photon-mediated ionization matters most for hard spectra that place appreciable particle number in the tens-of-MeV window. For $p = 1.8$, the fixed band lowers $\overline{W}$ by a factor of about three; for $p \geq 2.5$, low-energy electrons dominate both integrals and the result is unchanged. Even after adding the productive band, energy injected far above the plateau raises the numerator without proportionally increasing the yield.

If a source model supplies a comoving escaped-electron energy density $u_{e,\text{CR}}$ with a specified spectrum, the no-recombination ionization density is $n_{\text{ion}} = u_{e,\text{CR}}/\overline{W}$. Turning this into a global ionized fraction requires an evolving source history, recombinations, H/He chemistry and spatial overlap. Existing hadronic heating normalizations cannot be transferred directly to the electron component because the acceleration efficiency and escape fraction differ [Leite et al., 2017, Ruszkowski and Pfrommer, 2023].

8 Discussion and limitations

The calculation establishes two productive leptonic windows. Near keV energies, direct H collisions generate the familiar neutral-gas cascade. Near tens of MeV, IC photons inside the adopted absorption band produce a second cascade with comparable W . Between these windows the IC spectrum is only partly ionizing, and above the second window additional radiated energy moves beyond the assumed absorption ceiling [Blumenthal and Gould, 1970, Slatyer et al., 2009].

The magnitude of the new branch is conditional on the approved photon rule. One absorption per 13.6 eV–1 keV photon is a transparent limiting estimate, not a self-consistent opacity solution. A full treatment must redshift photons, distinguish absorption from Compton scattering, evolve the neutral fraction, include helium, and follow delayed deposition across the changing IGM [Kanzaki and Kawasaki, 2008, Slatyer, 2016]. The 13–37 per cent change produced by the interaction-horizon ceiling quantifies only one part of that uncertainty.

The homogeneous fixed- x_e approximation is likewise restrictive. Increasing x_e strengthens Coulomb heating and weakens the ionization share of low-energy secondaries; helium adds excitation, ionization and opacity channels [Furlanetto and Stoeve, 2010]. Recombinations make one ionization per H atom insufficient for sustained reionization, while source clustering and diffusion control whether deposition is local or volume-filling [Robertson, 2022, Jana and Nath, 2018].

Finally, the calculation begins after electron escape. The global impact may be small even when the per-particle yield is high if host columns stop sub-relativistic electrons or if accelerators place little power in either productive window. Conversely, a soft escaped spectrum or a component near tens of MeV can exploit the minima in Fig. 4. Source-specific escape and acceleration models are therefore the next necessary layer [Park et al., 2015, Ruszkowski and Pfrommer, 2023].

9 Conclusions

We have combined cosmological CR-electron cooling, a recursive electron-impact cascade and the differential IC photon spectrum. The main results are:

1. The direct cascade has a minimum $W \simeq 35.4 \text{ eV}$ near 1 keV and saturates at $(0.8\text{--}1.2) \times 10^4$ ionizations per injected electron, consistent with neutral-gas deposition calculations [Shull and van Steenberg, 1985, Furlanetto and Stoeve, 2010].
2. Under one absorption per 13.6 eV–1 keV photon, IC emission creates a second minimum $W \simeq 39 \text{ eV}$ near $(5\text{--}9) \times 10^7 \text{ eV}$ and raises the plateau to 2.9×10^6 ($z_i = 20$) or 4.0×10^6 ($z_i = 10$).
3. A redshift-dependent interaction-horizon ceiling increases those plateaux to 3.9×10^6 and 4.5×10^6 , demonstrating a 13–37 per cent opacity sensitivity but not replacing radiative transfer [Karzas and Latimer, 1961, Slatyer, 2016].
4. The photon branch reduces $\overline{W}$ most strongly for hard injection spectra; soft spectra remain controlled by the keV collisional window.
5. Path length is only an upper limit on displacement. A global prediction requires source escape, magnetic transport, recombinations and photon transfer [Jana and Nath, 2018, Ruszkowski and Pfrommer, 2023].

Code, algebra and reproducibility

The common loss model, direct cascade, photon cascade and figures are in `igm_losses.py`, `cascade_traj.py`, `photon_cascade.py`, `make_paper_figures.py` and `make_short_paper_figure.py`. Symbolic checks are in `verify_manuscript_algebra.py`; numerical text-table-figure regressions are in `verify_manuscript_consistency.py`.

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